

Random Versus Blocked Practice to Enhance Mental Representation in Golf Putting

Perceptual and Motor Skills

2017, Vol. 124(3) 674–688

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DOI: 10.1177/0031512517704106

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Abstract

The purpose of this study was to investigate changes in mental representation from either random or blocked practice when engaged in golf putting. Thirty participants were randomly assigned to random practice, blocked practice, and no-practice groups. First, we measured novice golfers' initial mental representation levels and required them to perform 18 putting trials as a pre-test. We then asked random and blocked groups to practice in accordance with their group assignment for six consecutive days (10 blocks each day, 18 trials each). A week after the last practice session, we re-measured all participants' final mental representation levels and required them to perform 18 putting trials to evaluate learning retention through practice. While those engaged in the random practice method putted more poorly during acquisition (i.e., practice) than those in blocked practice, the random practice group experienced more accurate retention during the final putting trials, and they showed a more structured mental representation than those in blocked practice, one that was more similar to that of skilled golfers. These results support the acquisition of a rich mental representation through random versus blocked practice.

Keywords

contextual interference, mental representation, random practice, blocked practice, structural dimensional analysis of mental representation

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Introduction

Contextual interference plays an important role in maximizing human motor learning (Shea & Morgan, 1979). Such interference is produced when several tasks must be practiced during acquisition. If practice of task variations is accomplished in large, isolated groups, then this blocked practice typically culminates in weaker learning; if practice with task variations is interspersed randomly during acquisition, then these random practice trials often culminate in better learning (Shea & Morgan, 1979). Many prior studies have attested to the utility of random practice with high contextual interference across varied tasks and background conditions (Brady, 1998; Magill & Hall, 1990).

Several explanatory theories for the contextual interference effect have been suggested since Shea and Morgan's (1979) original findings, each one emphasizing the learner's cognitive operations. Among these, hypotheses of elaboration and distinctiveness and of forgetting and reconstruction have been particularly popular. The elaboration and distinctiveness theory argues that interspersed variety in random practice provides an opportunity for the learner to compare and contrast various conditions and thereby develop a rich mental representation of task variety, creating more distinctive and detailed memories (Lin, Fisher, Winstein, Wu, & Gordon, 2008; Shea & Morgan, 1979). Alternatively, according to forgetting and reconstruction theory (Lee & Magill, 1983, 1985), practicing in random conditions forces the learner to remove the previous executed pattern from his/her working memory in each new attempt to plan and execute successive practice trials. Because each earlier pattern is dumped and superseded by successive new patterns of planning and execution, no clear pattern is immediately available and new patterns must be continuously reconstructed. This forgetting and reconstruction leads to a stronger memory representation than constant, repeated practice with a single movement pattern that is maintained in working memory across an entire series of blocked trials (Lee & Magill, 1983, 1985). Magill and Hall (1990) argued that creating interference by modifying a single parameter in the same class of a generalized motor program (GMP) might not create sufficient contextual interference to facilitate retention or transfer performance. Rather, action plans must undergo complex reconstruction during high contextual interference in more effortful information processing to create stronger mental representations of the required movements of the new task (Lee & Magill, 1983). The final action plan consists of an appropriate GMP with various parameters added to it (Magill & Hall, 1990). Therefore, in blocked practice, reconstruction by singular parameter modifications introduced one at a time requires only parameter reconstruction and leads to less effortful processing. Of note, however, some studies have shown the contextual interference effect via these parameter modifications (Sekiya, Magill, & Anderson, 1996; Sekiya, Magill, Sidaway, & Anderson, 1994; Wulf & Lee, 1993).

Although each of these two views explains the effect of contextual interference from different perspectives, both theories endorse random practice as a path toward stronger memory representation of the task than blocked practice; and both theories explain retention benefits by stronger mental representation of the task at hand. Although many studies have addressed contextual interference benefits (Schmidt & Lee, 1988), the authors are unaware of any specific empirical support for stronger memory representation in random versus blocked practice.

A technique called Structural Dimensional Analysis of Mental Representation (SDA-M) measures participants' mental representation based on non-declarative knowledge (Schack, 2012). Generally, previous studies of the structure of mental representation in complex actions have employed one of two methods: (1) those derived from response behaviors and (2) those derived from reaction times (Schack, 2012). While the first approach has been used to study the order formation in Long-Term Memory (LTM), the second has been applied to determining chunk structures in working memory (Schack, 2004). The mental representation of action in LTM has been studied through participant interview, paper-and-pencil tests, and specific forms of protocol analysis (Hodges, Huys, & Starkes, 2007). Thus, most of the many studies of knowledge-based representation in LTM have been non-experimental, focused on participants' explicit knowledge, and beset with objectivity and reliability problems (Thomas & Thomas, 1994). Yet, knowledge that is functionally relevant for motor control and the organization of actions may be distinguished from the participant awareness that merely accompanies these actions or justifies them retrospectively. We cannot assume that self-reported knowledge, even from top performers (experts), is the same as knowledge that was actually responsible for their performance (Schack, 2004, 2012). As an alternative technique, SDA-M measures individuals' mental structure based on nondeclarative knowledge and relies on familiarity with cognitive action architecture to study mental representation of actions in LTM (Schack, 2004, 2012). Schack (2004) believes that movements are well-structured concepts, hierarchically stored in LTM that can be recalled at any time and regenerated. This assumption forms the basis of the SDA-M measurement approach in which cognitive action architecture contributes to a mental representation of high-level motor skills organized within a hierarchical memory structure consisting of Basic Action Concepts (BACs; Schack, 2004). BACs denote the cognitive compilation of body postures along with sensory consequences that are functionally related to the attainment of action goals. From such an action architecture perspective, mental representations can be specified by well-integrated networks of these action concepts that serve as tools to facilitate motor system control (Frank, Land, & Schack, 2013). The storage of BACs in LTM can then be assessed by SDA-M.

SDA-M has been used to measure the mental representation of actions involved in the tennis serve (Schack & Mechsner, 2006), dance (Bläsing, Tenenbaum, & Schack, 2009), judo (Weigelt, Ahlmeyer, Lex, & Schack, 2011),

volleyball (Velentzas, Heinen, Tenenbaum, & Schack, 2010), and hand actions (Stöckel, Hughes, & Schack, 2012). Although most studies have examined the different representations experienced by experts and novices, there have also been studies investigating changes in mental representation resulting from development or practice. For example, Stöckel et al. (2012) evaluated children's development of mental representation for grasping postures, comparing 7-, 8-, and 9-year-old children. They found that 9-year-old children were more likely to plan their movements to end in comfortable postures and to have distinct representational structures of certain grasp postures, compared with 7- and 8-year-old children. In addition, Frank et al. (2013) used this method to show that, following putting practice, novice participants exhibited mental representation structure that was more similar to that of skilled golfers. Thus, SDA-M can be used to assess changes in mental representation of motor actions that result from practice and experience.

In the context of this prior literature, we sought to examine whether random practice, compared with blocked practice in learning to putt, would lead to a stronger mental representation in participants' memories, and whether those experiencing random practice would show mental representation closer to that of skilled golfers than blocked or control groups. Our hypotheses were as follows:

Hypothesis 1. Based on prior random versus blocked practice research (Sekiya et al., 1996; Sekiya et al., 1994; Wulf & Lee, 1993), we hypothesized that random group participants would perform more poorly during acquisition but show better retention performance following practice than blocked group participants.

Hypothesis 2. Based on prior practice related mental representation research (Frank et al., 2013), and on theoretical views of contextual interference (Lee & Magill, 1983; Shea & Morgan, 1979), we predicted that, after practice, mental representation of random group participants would be more structured and more similar to skilled performers than that of blocked group participants.

Method

Participants

Thirty male student participants (mean age = 27.4 ± 4.6 years) were randomly assigned to (1) random practice, (2) blocked practice, or (3) a no-practice control group. All participants were right-handed, had no previous golf putting experience, and signed informed consent forms. All methods used in the study were approved by the Ethics Committee of Ferdowsi University of Mashhad.

Tools and Tasks

The experiment used a synthetic 9×4 meter putting green (STI Pro Putt). All participants used a standard right-handed putter and standard golf balls. For creating contextual interference, three circular targets with a diameter of 4 cm were placed at different distances of 1.22, 2.44, and 3.66 meters. SDA-M was used to assess the participants' mental representation.

Structural Dimensional Analysis of Mental Representation

To run SDA-M, before analysis, golf putting BACs (providing the basis for participants' responses and subsequent analysis) were chosen by reference to previous research (Frank et al., 2013) that used objective methods to determine them. The following 16 BACs were selected: (a) shoulders parallel to target line, (b) align club face square to target line, (c) grip check, (d) look to the hole, (e) rotate shoulders away from the ball, (f) keep arms-shoulder triangle, (g) smooth transition, (h) rotate shoulders toward the ball, (i) accelerate club, (j) impact with the ball, (k) club face square to target line at impact, (l) follow-through, (m) rotate shoulders through the ball, (n) decelerate club, (o) direct club head to planned position, and (p) look to the outcome. From biomechanical and functional points of view, each of the 16 BACs can be attributed to a particular motor phase: preparation (BAC 1–4), backswing (BAC 5–7), forward swing (BAC 8–11), and attenuation (BAC 12–16).

The SDA-M involves four steps (Schack, 2012): (1) a special split procedure involving a multiple sorting task delivers a distance scaling between the BACs of a suitably pre-determined set; (2) a hierarchical cluster analysis is used to outline the structure of the given set of BACs; (3) a factor analysis reveals the dimensions in this structured set of BACs; and (4) the cluster solutions are tested for invariance within and between groups. The structure of mental representation was assessed using a splitting task in order to learn about the distance between BACs in memory; this splitting task was performed in front of a computer with the screen displaying the BACs of the golf putt. In detail, the splitting task proceeds as follows: a BAC is permanently displayed on the screen (anchor concept) while other BACs are presented successively in a random order. Participants are asked to decide, one after another, whether a given BAC is related to the anchor concept or not during movement execution. Once a given list of BACs is finished, the next BAC serves as an anchor concept, and the procedure continues. The splitting task ends after each BAC has been compared with the remaining BACs in the list.

Procedure

Pre-test. To get familiar with the task, all participants watched a video in which a skilled player performed the putting task. Then, the examiner made the

participants familiar with the splitting task. At first, the participants were shown a list of 16 BACs of golf putting in a random order. The examiner explained the meaning of each BAC to the participants in order to ensure comprehension. Then, the instructions shown on the screen were explained to the participants. Specifically, they were instructed to decide whether the BACs were related to one another or not during movement execution. After these explanations, the participants performed the splitting task in order to determine their initial mental representation structures of the putting task. Then, the participants completed 18 putting trials as a pretest. The trials' order was random with the constraint that no target should be repeated twice in a row and each target should be repeated six times. The participants attempted to putt the ball, so it landed directly on the specified target and they were not given any instructions about how to putt the ball.

Acquisition phase. During acquisition, the participants practiced based on their grouping. The random group performed 10 blocks (18 trials each) in random order every day, while the blocked group completed 10 blocks (18 trials each) in blocked order every day (blocked group practiced one target in a day, resulting in two consecutive practice session for each target). To reduce practice order effects, half of the participants in the blocked group performed the task from the farthest target to the nearest one, and the other half practiced from the nearest target to the farthest one. The participants practiced in the task for six consecutive days, during which time the control group did not practice.

Retention. One week after the last acquisition practice session, all participants performed the splitting task again to specify their final structure of mental representation of the putting task. Then, they performed 18 putting trials (in random order) as a putting retention test.

Performance Data Analysis

To assess the participants' putting performance (accuracy), the ball's resting distance from the target after each putt was measured in centimeters (taken from the edge of the target). The dependent measure for putting performance was then determined by averaging putting accuracy data for each test and day of practice. A one-way analysis of variance (ANOVA) was used to compare group performance at pretest. To compare the practice performance of the random and blocked groups during acquisition, practice data were analyzed in a 2 (groups; random and blocked) $\times$ 6 (days of practice) mixed design ANOVA with repeated measures on the last factor (note that since the control group did not practice during acquisition, it was excluded from practice analysis). Finally, in the retention phase, data were again analyzed in a one-way ANOVA with the control group included.

Structure of Mental Representation

The structure of mental representation was determined by calculating mean group dendrograms via cluster analysis in accordance with the four-step process described earlier. For all cluster analyses conducted, an alpha-level of $\alpha = .05$ was chosen, resulting in a critical value, $d_{crit} = 3.41$. Links between BACs higher than this value were considered statistically irrelevant while BACs linked under this critical value were considered statistically relevant. To compare the differences between cluster solutions, analyses of invariance were performed. Two cluster solutions were variant (i.e., significantly different) for $\lambda < 0.68$, while two cluster solutions were invariant for $\lambda > 0.68$ (Schack, 2012). Mental representations of two skilled performers (putters with at least 10 years of golf experience ranked within the top 10 Iranian golfers) served as a reference structure which reflected well the four movement phases (i.e., preparation, backswing, forward swing, and attenuation) of the putting action. To test the similarity of the mental representation of these experimental groups and that of the skilled performers, the Adjusted Rank Index (ARI) was used (Santos & Embrechts, 2009). This index serves as an index of similarity in a range of (1, -1). On this scale, the value (-1) indicates that two cluster solutions are different, and the value (1) indicates that two cluster solutions are the same. Indices between these extremes rank similarly between two cluster solutions.

Results

Performance

Figure 1 shows the group performances at pretest, acquisition, and retention. The results of one-way ANOVA showed no significant group differences at pretest ($F < 1$).

For the acquisition phase (Figure 1), the performance of both practice groups improved over time. There were significant main effects of group, $F(1,18) = 59.93$, $p < .0001$, $\eta_p^2 = 0.76$, and practice days, $F(5,90) = 47.56$, $p < .0001$, $\eta_p^2 = 0.72$. An interaction effect of group and practice days was also significant, $F(5,90) = 5.91$, $p < .001$, $\eta_p^2 = 0.24$, and group mean comparisons indicated that the random practice group performed more poorly than the blocked practice group during acquisition phase (means, random = 48.35, block = 38.33). For the main effect of practice days, the LSD post hoc test showed no significance difference between practice days after the third day of practice, all $p > .05$ (however, there was a significant difference between the third and sixth days, $p < .05$), but differences between the first, second, and third days were significant, all $p < .05$. Group mean comparisons showed performance improvement in both groups during practice days (Figure 1). For the interaction effect, post hoc test analyses revealed a significant group difference at the

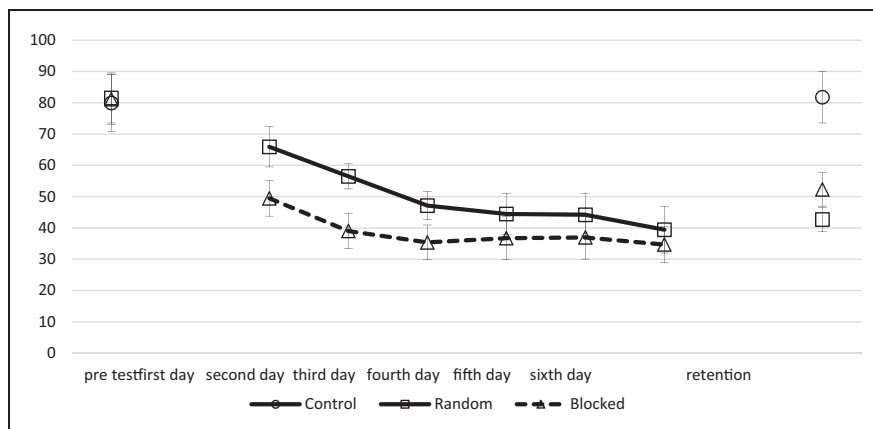


Figure 1. Performance of groups at different stages

beginning of practice; however, there was no significant difference at the sixth day, $p > .05$.

For the retention test, there was a significant group difference, $F(2,29) = 53.12$, $p < .0001$, and LSD post hoc testing showed a significantly better performance by both experimental groups in comparison to the control group, all $p < .05$ (means, random = 43.70, blocked = 52.32, and control = 81.78), and a significantly better performance by the random practice group, compared with the blocked practice group, $p < 0.05$.

Mental Representation Structure

Cluster analysis specified no clustering in mean group dendrograms at pretest ($d_{crit} = 3.41$).

Although there was no significant group structure difference at pretest, one week after practice, mean group dendrograms showed an increase in the number of functional clusters for both practice groups (Figure 2). For both the random and blocked groups, statistical analysis of invariance showed significant pretest posttest differences in the structures of mental representation (all $\lambda < 0.68$); the control group showed no significant pretest posttest difference in the structure of mental representation ($\lambda > 0.68$).

For the random practice group, BACs were clustered into four functional units: (1) Preparation phase with BAC1 (shoulders parallel to target line), BAC2 (align club face square to target line), BAC3 (grip check), and BAC4 (look to the hole); (2) Forward swing with BAC8 (rotate shoulders toward the ball) and BAC9 (accelerate club); (3) Club head-ball impact with BAC10 (impact with the ball) and BAC11 (club face to target line at impact); and (4) Attenuation

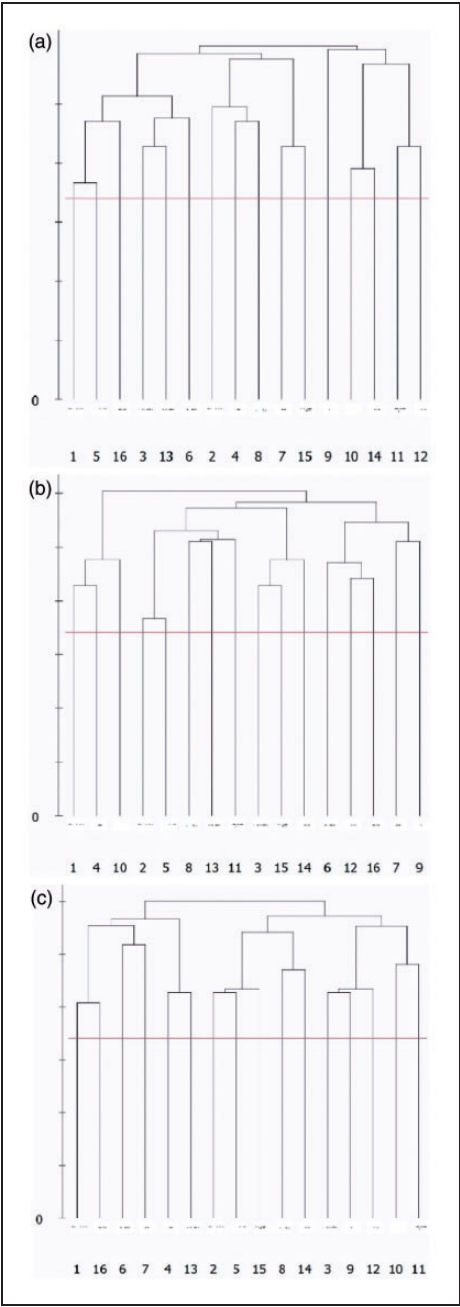


Figure 2. Mean group dendrograms of (a) the control group, (b) the blocked group, and (c) the random group at pre-test.

phase with BAC13 (rotate shoulders through the ball), BAC14 (decelerate club), BAC15 (direct club head to planned position), and BAC16 (look to the outcome). However, for the blocked group, BACs were clustered into only two functional units: (1) Preparation phase with BAC1 (shoulders parallel to target line), BAC2 (align club face square to target line), BAC3 (grip check), and BAC4 (look to the hole); and (2) Attenuation with BAC14 (decelerate club) and BAC15 (direct club head to planned position).

To assess the degree of functional adaptation of these mental representations, the mental representation of each group (random, blocked, and no practice control) was compared with that of skilled performers ($n=2$). The ARI showed that, after practice, the mental representation of both practice groups (random and blocked) was more similar to that of skilled golfers than that of the control group. The random practice group showed particular progress toward the structure of mental representation of skilled performers, with a large increase in ARI from pretest to posttest (from 0.06 to 0.72). The blocked practice group also showed a more modest ARI increase (from 0.09 to 0.27), indicating that, after practice, mean dendograms of the blocked group had also become more similar to the reference dendogram from experienced golfers. In contrast, the control group's structure showed only a slight trend toward the reference structure over time, with almost no ARI increase (from 0.03 to 0.06). After practice, the mental representation of the random group was most similar to that of skilled performers (ARI for random group = 0.72, ARI for blocked group = 0.27).

Although the random practice group performed more poorly during acquisition, they performed more accurately during retention than blocked practice group. Thus, these results supported Hypothesis 1. Also, as the random group showed more structured changes in mental representation compared with the blocked group, and their mental representation was more similar to that of skilled performers than the blocked group, Hypothesis 2 was also supported.

Discussion

It has been well established that high contextual interference associated with practicing motor tasks in random order leads to weak acquisition performance but better retention and transfer, compared with low contextual interference associated with practicing in blocked order, and this effect has been called the contextual interference effect (Magill & Hall, 1990; Shea & Morgan, 1979). Different theories ascribe better learning by random versus blocked practice to the development of a stronger mental representation for the task (Lee & Magill, 1983; Shea & Morgan, 1979). This study examined changes in the structure of mental representation due to random or blocked practice of golf putting. Our results showed confirmed Shea and Morgan's (1979) context interference effect

in that the random practice group putted more poorly during practice acquisition than the blocked practice group but putted better than the blocked group at retention testing. Although it has been argued that parameter modifications of the same GMP do not create sufficient interference to require the kind of additional processing during acquisition that facilitates retention or transfer performance, in line with previous findings (Sekiya et al., 1996; Sekiya et al., 1994; Wulf & Lee, 1993), our findings showed that changes in the parameters of a class of action which require the same GMP could create the contextual interference effect.

Along with better putting at retention testing for the random versus blocked practice participants, we also found that the mental representation of putting in the random practice group was more structured, compared with the blocked practice group. Figures 2 and 3 show that the random group had structured the BACs into four clusters (related to preparation, forward swing, hitting the ball, and attenuation), whereas the blocked group structured the BACs into only two clusters (related to preparation and attenuation). Moreover, comparing the structure of mental representation of each group with skilled golfers, the structure of mental representation of putting for the random practice group was more similar to that of skilled performers than was the blocked practice group, even though the blocked group also outperformed a control (no-practice) group in both performance precision and mental representation changes.

The results of the current study extended previous findings in the fields of contextual interference and mental representation. Two different theories of elaboration and distinctiveness (Shea & Morgan, 1979) and forgetting and reconstruction (Lee & Magill, 1983) have been offered to explain the contextual interference effect. These two theories differ as to the mechanism of contextual interference, but they overlap in affirming that random practice creates a strong mental representation. Our findings support this link since, in this study, random practice participants not only gave a more accurate performance at the retention test compared with the block group but also showed a more structured mental representation than the blocked practice group and a mental representation that was more similar to that of skilled performers. Given the functional relationship between mental representation and motor performance (Schack, 2012; Velentzas et al., 2010; Weigelt et al., 2011) and the similarity of mental representation between random practice participants and skilled golfers, it is likely that this more structured mental representation helped participants in the random group enhance their putting performance at the retention test.

Early mental representation researchers showed that higher order mental representation is associated with high skill level while low-order long-term memory formation is associated with low skill level (Schack & Mechsner, 2006). Additionally, researchers have recently shown that practice leads to a functional adjustment in the structure of mental representation (Frank et al., 2013) suggesting that different practice methods can lead to different functional

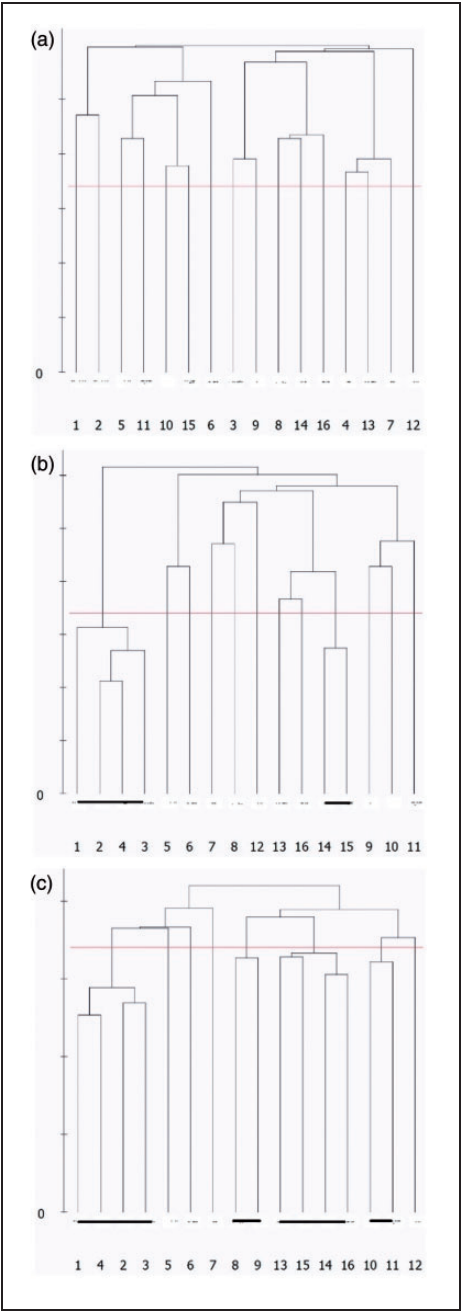


Figure 3. Mean group dendrograms of (a) the control group, (b) the blocked group, and (c) the random group at post-test.

changes in the structure of mental representation of motor tasks. This study has now extended that research vein by showing that random practice, compared with blocked practice, does, in fact, result in a more structured mental representation of golf putting.

While contextual interference effect research has mainly focused on outwardly apparent performance effects while ignoring covert cognitive effects, this study merged mental representation and contextual interference research threads to address underlying cognitive changes associated with different practice methods. Our data suggest that random versus blocked practice facilitates both a more structured mental representation and an improved putting performance during retention testing. Of note, however, in the early stages of skill acquisition, motor learning seems to be associated with forming a mental representation of action-related knowledge in LTM, and it seems that such order formation is better facilitated via random than blocked practice.

This study provides the first direct evidence of improved mental representation changes associated with contextual interference. There are some limitations in this study that should be considered in future research. Although, in this study contextual interference was created by parameter modification, this effect also can be created by GMP modifications. Furthermore, in this study, contextual interference was addressed using random and blocked practice; nevertheless, different degrees of contextual interference might be created by serial, random, and blocked practice (Magill & Anderson, 2007). According to these limitations, future research might compare functional mental representation changes of the performers under contextual interference using different movement patterns and parameters. Since it has been argued that contextual interference created by changing the parameters of the same movement pattern (like this study) is low compared with interference created by different movement patterns (Lee, Wulf, & Schmidt, 1992; Magill & Hall, 1990), it is possible that representational changes by GMP modifications would be greater than representational changes by parameter modifications. Hence, future studies should examine structural changes in mental representation that may result from more complex contextual interference using both GMP and parameter modifications and different degrees of contextual interference from serial, random, or blocked practice (Magill & Anderson, 2007).

Acknowledgment

The author(s) thank Professor Thomas Schack for providing help with SDA-M method.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

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